Variational Autoencoders

March 21, 2025

Agenda

Bayesian Statistics

- Bayesian statistics offers a probabilistic approach to inference.
- Uses prior knowledge and updates beliefs with observed data based on Bayes' Theorem:

$$p(\theta|\mathcal{D}) = \frac{p(\mathcal{D}|\theta)p(\theta)}{p(\mathcal{D})} \propto p(\mathcal{D}|\theta)p(\theta)$$

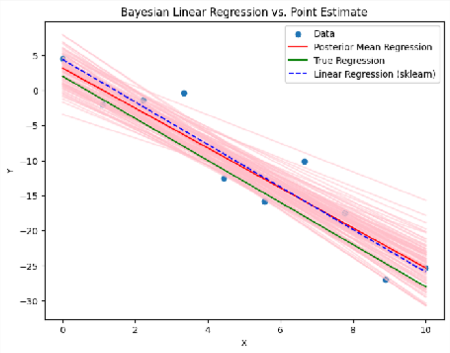
Where:

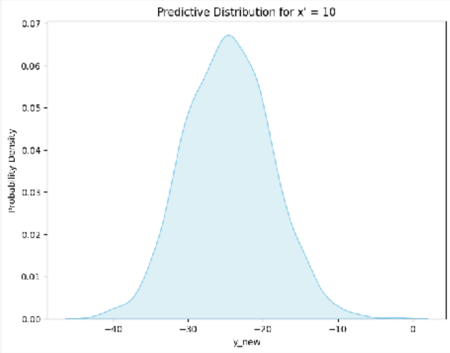
- $P(\theta|\mathcal{D})$ - Posterior probability (updated belief)
- $P(\mathcal{D}|\theta)$ - Likelihood (data given parameter)
- $P(\theta)$ - Prior probability (initial belief)
- $P(\mathcal{D})$ - Evidence/marginal probability (normalization factor)

Use the posterior to make predictions about future data $\mathbf{x}'$
Posterior predictive distribution:

$$p(\mathbf{x} = \mathbf{x}'|\mathcal{D}) = \int p(\mathbf{x} = \mathbf{x}'|\theta)p(\theta|\mathcal{D})d\theta$$

Comparison between Frequentist & Bayesian Regression



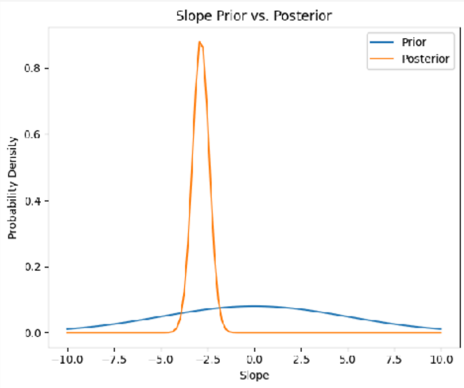


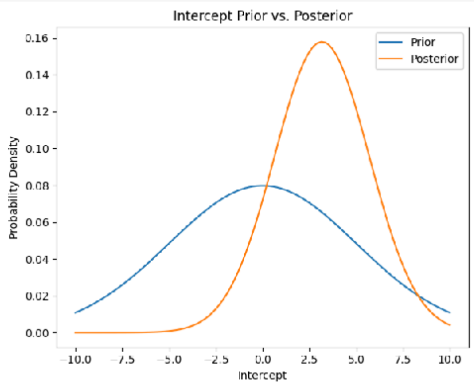
$y = -3x + 2$

Posterior predictive distribution (in conditional or supervised learning case):

$$p(\mathbf{y}|\mathbf{x} = \mathbf{x}', \mathcal{D}) = \int p(\mathbf{y}|\mathbf{x} = \mathbf{x}', \theta) p(\theta|\mathcal{D}) d\theta$$

Prior and posterior distributions over parameters θ





Maximum Likelihood Estimate

Toy Example

A fair coin is tossed **3 times**, and we observe **3 heads**.
Compute **Likelihood Estimation (MLE)** and **Bayesian Inference**.

$$P(D|\theta) = \theta^3(1-\theta)^0 = \theta^3$$

MLE Estimate:

$$\hat{\theta}_{\text{MLE}} = \arg\max_{\theta} \theta^3 = 1$$

Issue: Overconfident estimation (not ideal for small data).

Bayesian Inference

Bayesian Inference
Bayes' Theorem:

$$P(\theta|\mathcal{D}) = \frac{P(\mathcal{D}|\theta)P(\theta)}{P(\mathcal{D})}$$

Prior: Assume a uniform prior : $P(\theta) = \text{Beta}(1,1)$

$$\textbf{Posterior:} P(\theta|\mathcal{D}) = \text{Beta}(1+3, 1+3-3) = \text{Beta}(4,1)$$

Posterior Mean:

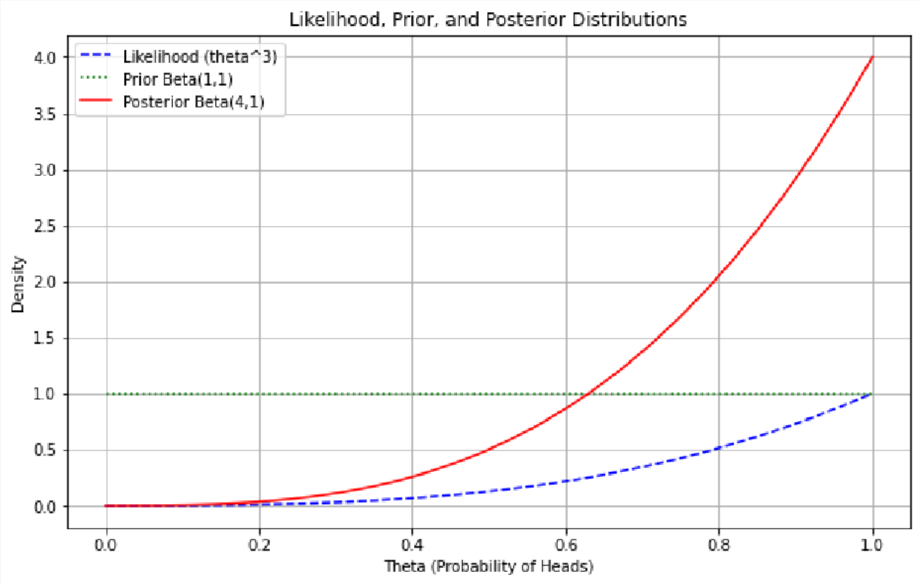
$$E[\theta|\mathcal{D}] = \frac{\alpha}{\alpha+\beta} = \frac{4}{4+1} = 0.8$$

Bayesian Estimate
Comparing the estimates:

- MLE Estimate: $\theta = 1$ and Bayesian Estimate: $\theta = 0.8$

Thus, the Bayesian solution is more robust to small datasets, as it incorporates prior information.

How the coin-example distributions look like



There are other benefits too of Bayesian Analysis

- Incorporate Prior Knowledge
- Quantify uncertainty through posterior distributions enabling to assess the credibility of your estimates and make more informed decisions.
- It is ideal for online learning, where you update your beliefs as new data becomes available
- Provide more reliable inferences with small sample sizes, especially when informative priors are used.

However, the Bayesian methods can also be computationally intensive and require careful selection of prior distributions.

Latent Variable Model (LVM)

- A statistical model where some variables are not directly observed but are inferred from observed data. These unobserved variables are called latent variables, typically denoted by Z.
- The aim is to model the relationship between X and Z to better capture hidden structures and model complex data distribution.
- The directed graphical model would then represent a joint distribution $p_\theta(\mathbf{x}, \mathbf{z})$ over both the observed variables $\mathbf{x}$ and the latent variables $\mathbf{z}$.
- The marginal distribution over the observed variables $p_\theta(\mathbf{x})$, is given by:

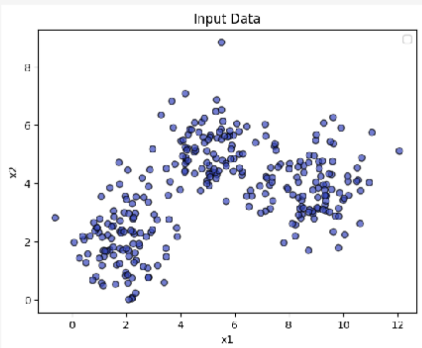
$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}, \mathbf{z})\, d\mathbf{z} \tag{1}$$

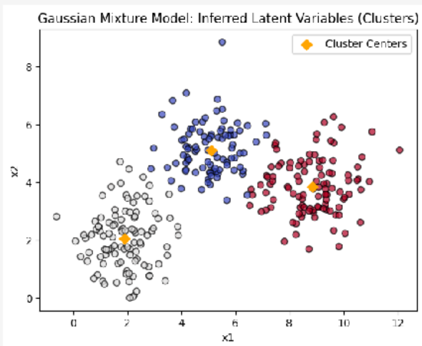
Latent Variable Model (LVM)

- The main challenge of maximum likelihood estimation in LVMs is that the marginal probability of data under the model is typically intractable.
- This is due to the integral that appears in computing the marginal likelihood (or model evidence), $p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}, \mathbf{z})\, d\mathbf{z}$.
- Due to this intractability, we cannot differentiate it w.r.t. its parameters and optimize it, as we can with fully observed models.
- The intractability of $p_\theta(\mathbf{x})$, is related to the intractability of the posterior distribution $p_\theta(\mathbf{z}|\mathbf{x})$. However, the joint distribution $p_\theta(\mathbf{x}, \mathbf{z})$ is efficient to compute.

$$p_\theta(\mathbf{z}|\mathbf{x}) = \frac{p_\theta(\mathbf{x}, \mathbf{z})}{p_\theta(\mathbf{x})} = \frac{p_\theta(\mathbf{z}) p_\theta(\mathbf{x}|\mathbf{z})}{p_\theta(\mathbf{x})}$$

Example of Latent Variable Model





Variational Autoencoders (VAE)

- The goal is to learn the underlying probability distribution of the data
- A probabilistic version of a deterministic autoencoder – unlike standard autoencoders, VAEs do not learn a fixed latent code for each input, instead, they learn a probability distribution (e.g., Gaussian) over the latent space.
- Employs tools such as Variational Inference & KL Divergence
- VAEs are generative models and can create new samples.

VAE Architecture

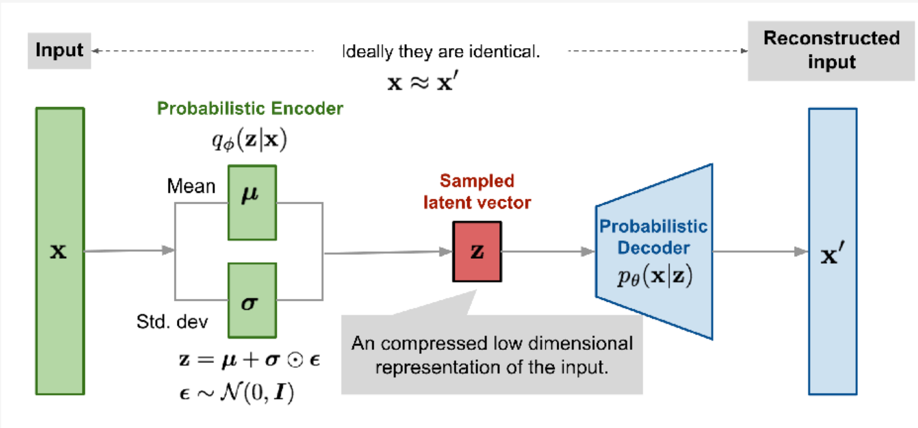


Image source [3]

Loss Function: Evidence Lower Bound (ELBO)

- The goal is to make $q_\phi(\mathbf{z}|\mathbf{x})$ close to the true posterior $p_\theta(\mathbf{z}|\mathbf{x})$
- For any choice of inference model $q_\phi(\mathbf{z}|\mathbf{x})$, including the choice of variational parameters ϕ, we have:

$$\log p_\theta(\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x})\right]$$

$$\vdots$$

$$\vdots$$

$$= \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log\left[\frac{p_\theta(\mathbf{x},\mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})}\right]\right]}_{\substack{=\mathcal{L}_{\theta,\phi}(\mathbf{x}) \\ \text{(ELBO)}}} + \underbrace{\mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log\left[\frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z}|\mathbf{x})}\right]\right]}_{=D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}|\mathbf{x}))}$$

Loss Function: Evidence Lower Bound (ELBO)

The ELBO, also known as *variational lower bound* is:

$$\mathcal{L}_{\theta,\phi}(\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x},\mathbf{z}) - \log q_\phi(\mathbf{z}|\mathbf{x})\right]$$

simplified further as:

$$\mathcal{L}_{\theta,\phi}(\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}|\mathbf{z})\right] - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}))$$

The last equation can be viewed as the expected log likelihood, and a regularizer, that penalizes the posterior from deviating too much from the prior.

ELBO: Reconstruction and Regularization

$$\mathcal{L}_{\theta,\phi}(\mathbf{x}) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})}\left[\log p_\theta(\mathbf{x}|\mathbf{z})\right] - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}))$$

In other words, the VAE loss function consists of two terms:

- **Reconstruction Loss**: $-\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x}|\mathbf{z})$
  - Measures how well the decoder can reconstruct the input from the latent code.
- **Regularization Term**: $D_{\text{KL}}(q_\phi(\mathbf{z}|\mathbf{x})\|p_\theta(\mathbf{z}))$
  - Ensures that the learned latent distribution is close to the prior distribution.

The Reparameterization Trick

- **Problem**: In Variational Autoencoders (VAEs), the loss function involves generating samples from $\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})$.
- **Challenge**: Sampling is a stochastic process, making backpropagation of gradients impossible. It is a non-differentiable operation. A sampled value is just a discrete realization and does not smoothly change with small changes in the parameters of the distribution (mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$).
- **Solution**: The reparameterization trick allows us to express the random variable $\mathbf{z}$ as a deterministic variable.

Reparameterization Trick

- Since sampling from $q_\phi(\mathbf{z}|\mathbf{x})$ is a stochastic process, preventing backpropagation, the reparameterization trick expresses the latent variable $\mathbf{z}$ deterministically:

$$\mathbf{z} = \mathcal{T}_\phi(\mathbf{x}, \epsilon)$$

- For a multivariate Gaussian $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}, \boldsymbol{\sigma}^2 \boldsymbol{I})$:

$$\mathbf{z} = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \epsilon, \text{ where } \epsilon \sim \mathcal{N}(0, \boldsymbol{I})$$

- This allows gradients to flow during training, making the VAE trainable.

Illustration of the reparameterization trick

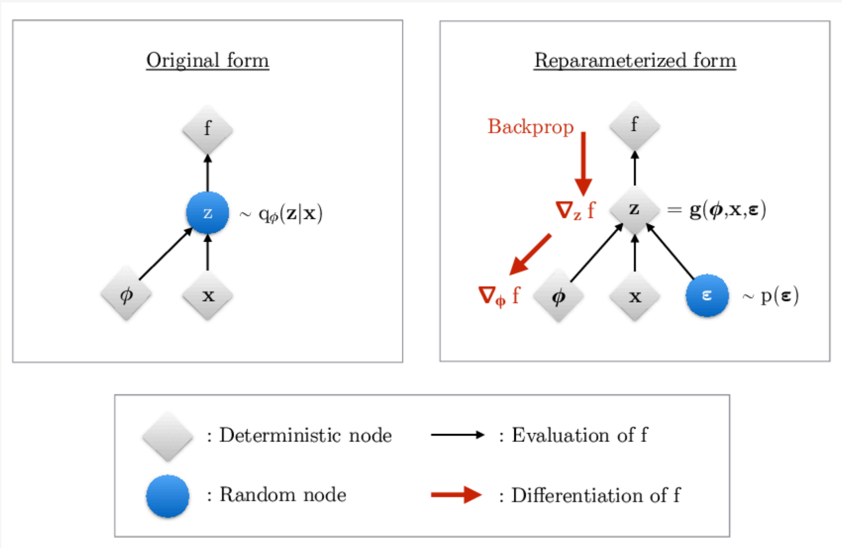


Image source [4]

Computation graph for VAEs

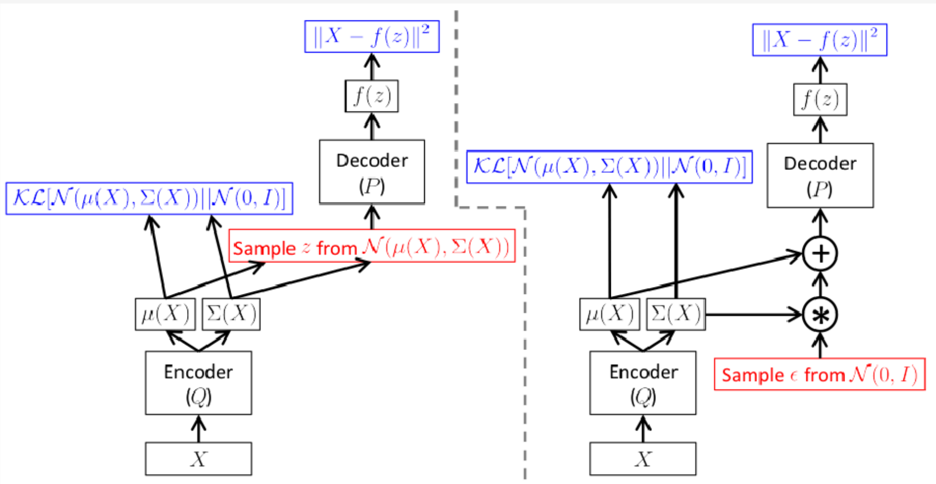


Image source [1]

Summary of VAE

- VAEs learn a probability distribution of the data by mapping inputs to a latent distribution.
- They consist of a probabilistic encoder and decoder.
- The model is trained using the ELBO loss function which includes a reconstruction term and a regularization term.
- The reparameterization trick is crucial for making the VAE trainable.

The behaviour of Forward and Reverse KL

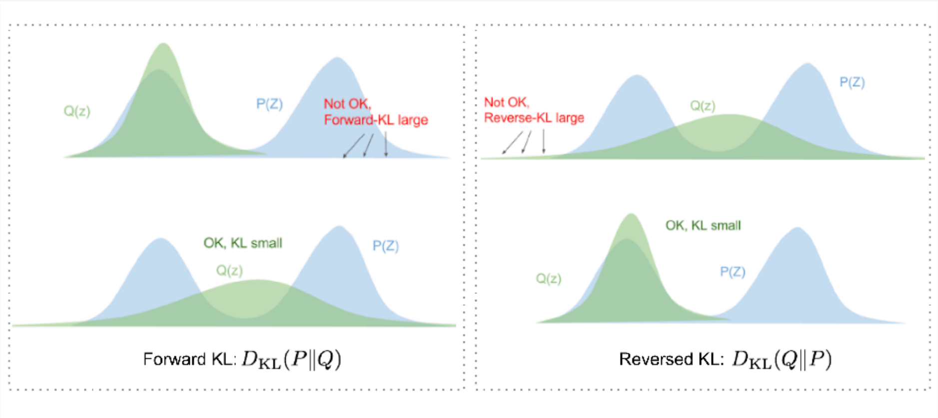


Image source [3]

- $Q(Z)$ will be zero-avoiding or mode-covering and will typically overestimate the support of $P(Z)$
- $Q(Z)$ will be zero-forcing or mode-seeking and will typically underestimate the support of $P(Z)$

References

1. Probabilistic Machine Learning: An Introduction by *Kevin P. Murphy*
2. Deep Learning - Foundations and Concepts by *Christopher M. Bishop*, *Hugh Bishop*
3. From Autoencoder to Beta-VAE `https://lilianweng.github.io/posts/2018-08-12-vae/`
4. An Introduction to Variational Autoencoders by *Diederik P Kingma* and *Max Welling* – Foundations and Trends® in Machine Learning.

Some of the images are sourced from the above-mentioned references